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Neoproterozoic palaeomagnetic directions in rocks from a key section of the Protogine Zone, southern Sweden

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SUMMARY

New palaeomagnetic data from the Protogine Zone (PZ) of southern Sweden are presented together with data from east and west of the zone. They confirm previous palaeomagnetic results from the PZ that indicate a difference in magnetic properties between rocks north and south of a section at 57°N, here called the Alvesta–Ljungby Palaeomagnetic Borderzone (ALPB). The results are divided into four groups: group I [declination (D) = 297°, inclination (I) = −78.5°, α_{95} = 2.4°, pole position 42.7°S, 220.9°E]; group II (D = 129.4°, I = 52.2°, α_{95} = 4.2°, pole position 9.4°S, 235.5°E); group III (D = 155.4°, I = 61.3°, α_{95} = 4.9°, pole position 11.6°S, 211.2°E); group IV (D = 125.5°, I = 35.4°, α_{95} = 5.4°, pole position 0.8°N, 244.1°E). A fifth, poorly defined, group has a mean direction D = 304.5°, I = 5.4°, α_{95} = 15.1° and pole position 21.3°N, 255.9°E. These palaeopoles fall on the *c.* 1050–900 Ma Sveconorwegian Loop of the Fennoscandian apparent polar wander path. Together with previously published data from southern Norway, the group I poles define the high-latitude apex of the loop at *c.* 950–930 Ma. The groups II–V palaeopoles are situated at low latitudes but it is difficult to place them in chronological order owing to a lack of age data and two possible interpretations, that is of an open or a closed loop.

The study shows that the part of the PZ extending south of the ALPB belongs to the Southwest Granulite Region and the rocks became remagnetized during the tectonic stacking and uplift processes that created that region. Dolerites from east of the PZ have primary magnetizations and the data indicate that they belong to the Blekinge–Dalarna dolerites, a suite of dykes east of the PZ with ages between 995 and 880 Ma. Dolerites west of the PZ yield low-latitude Sveconorwegian palaeopoles but it is not certain if their magnetizations are primary or are due to remagnetization during the Sveconorwegian Orogeny.

Key words: dolerite, Fennoscandia, Neoproterozoic, palaeomagnetism, Protogine Zone, remagnetization.

1 INTRODUCTION

The Protogine Zone (PZ) of southern Sweden (Fig. 1) defines a metamorphic boundary separating granulite–amphibolite facies rocks to the west and greenschist facies rocks to the east, with a metamorphic transition within the zone (e.g. Gorbatschev & Bogdanova 1993). The zone has been intruded by sets of dolerites ranging in age from *c.* 1550–900 Ma and by syenites dated at *c.* 1200 Ma. The differences in metamorphic grade and the presence of intrusions make rocks in the zone prime targets for palaeomagnetic study. Recent age determinations have enhanced the possibilities for using the results of such studies for defining the Early Neoproterozoic part of the Fennoscandian apparent polar wander path (APWP) and elucidating geological events within the PZ.

In a study by Bylund (1992) it was pointed out that the PZ can be divided palaeomagnetically into two parts, with the dividing line positioned across the zone at *c.* 57°N, in an east–west belt between the townships of Alvesta and Ljungby. For this study, the name Alvesta–Ljungby Palaeomagnetic Borderzone (ALPB) is used for this feature (Fig. 2). South of this zone all studied PZ rocks have single-component magnetization denoted by A, with mean D = 291°, I = −78°. North of this zone some A-directions were obtained at a few sites, but the dominant direction (B) is directed D = 129°, I = 40°. These magnetic components are found in both dolerites

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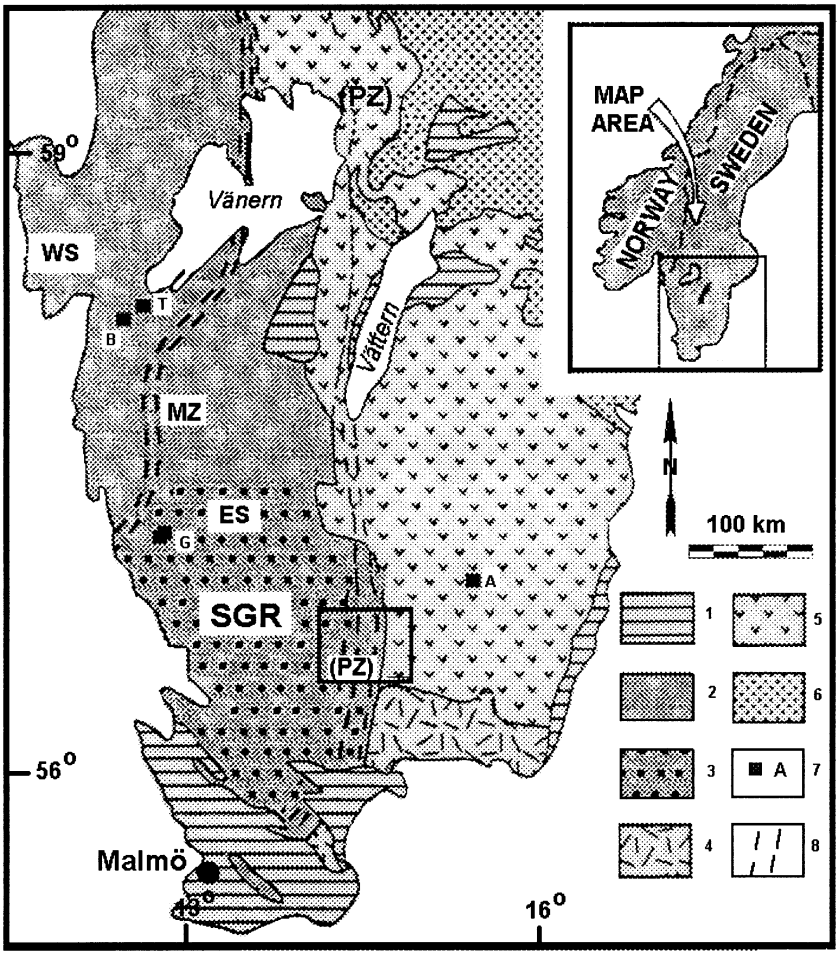


Figure 1. General geological map. 1: Phanerozoic sedimentary cover; 2: undifferentiated meta-igneous and metasedimentary rocks; 3: high pressure granulite facies and garnet amphibolite facies rocks (SGR); 4: Blekinge undifferentiated rocks; 5: Transcandinavian Igneous Belt (TIB); 6: Svecofennian rocks; 7: sampled sites outside the PZ (Å = Åseda, G = Gällared, B = Bredfjället, T = Trollhättan); 8: tectonic lineaments (PZ = Protogine Zone, MZ = Mylonite Zone). Box denotes area of Fig. 2.

and syenites and in their country rocks. Both groups give palaeopoles on the Neoproterozoic 'Sveconorwegian Loop' [as defined by Patchett & Bylund (1977)] of the Fennoscandian APWP. The A-group poles are situated at the apex of the loop at *c.* 45°S and 225°E, while the B-group poles cluster close to the equator at *c.* 230°E.

New radiometric ages have been obtained recently from amphibolites and granulites west of the PZ, approximately aligned with the ALPB (Wang *et al.* 1996b; Andersson 1996; Möller & Söderlund 1997). In this paper we present palaeomagnetic data from dolerites from within the ALPB and from dated rocks west of the PZ in order to improve the Sveconorwegian part of the Fennoscandian APWP (Elming *et al.* 1993). For comparison, we include new data from one dolerite at Åseda, east of the PZ, and from two dolerites situated west of the PZ (Fig. 1).

2 GEOLOGY AND FIELD SAMPLING

2.1 Geological outline

The geology of southwestern Sweden is characterized by tectonic zones that divide the region into different geological units. Important tectonic zones are the PZ and the Mylonite Zone (MZ, Fig. 1). The PZ forms the border between the Svecokarelian domain in eastern Sweden, dated mostly 1900–1650 Ma, and the Sveconorwegian domain in southwestern Scandinavia dated *c.* 1050–900 Ma. A belt of granites and porphyries, the Transscandinavian Igneous Belt (TIB) is juxtaposed between the Svecokarelian units and the PZ. The PZ itself is a 15–30 km wide north-trending zone of sheared and schistose gneisses of granitic and tonalitic composition, partly of TIB origin. Its borders in the study area are uncertain owing to a lack of outcrops and differing opinions about how lithologies and tectonic features are to be interpreted. There is a gradual increase in metamorphic grade across the PZ from greenschist facies rocks in the east to granulite facies rocks in the west (Johansson 1992). The PZ forms a metamorphic border between the little metamorphosed TIB and Svecokarelian rocks east of the zone, and the highly metamorphosed rocks to the west in the Sveconorwegian domain.

The area west of the PZ is dominated by gneisses belonging to the Southwest Swedish Gneiss Region. This is in turn divided by the MZ into an eastern segment between the MZ and the PZ and a western segment west of the MZ. The

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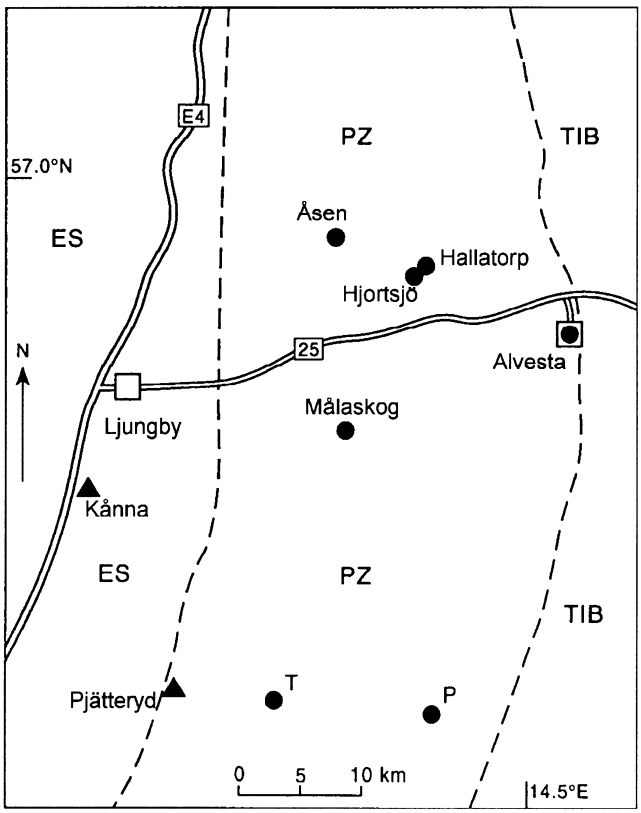


Figure 2. Sketch map of the sampled area between Alvesta and Ljungby. P and T are dolerite sites studied previously (Bylund 1992). Circles denote sampled dolerites, triangles sampled amphibolites and gneisses. PZ = Protogine Zone; TIB = Transscandinavian Igneous Belt; ES = Eastern segment.

southern part of the eastern segment is characterized by numerous occurrences of rocks of high metamorphic grade, which have undergone high-pressure granulite facies metamorphism (Möller *et al.* 1996, Fig. 1). This region has been named the Southwest Swedish Granulite Region (SGR; Johansson, Lindh & Möller 1991).

2.2 Age relations

The granitoids and volcanics in the TIB east of the study area yield ages between 1.81 and 1.77 Ga (Mansfeld 1996). East of the PZ a set of north-trending dolerites occur. They trend parallel to the PZ and are considered to have been intruded as a result of flexing due to uplift of the Sveconorwegian terrain (e.g. Patchett & Bylund 1977). This set of dolerites is referred to as the Blekinge Dalarna Dolerites (BDD). They yield whole-rock and mineral Rb–Sr ages between 995 and 880 Ma (Patchett 1978) and Sm–Nd ages between 935 and 844 Ma (Johansson & Johansson 1990). Within the PZ there are at least three sets of intrusions. The oldest is a set of dolerites that occur both within, and to the west of, the PZ, mainly in its northern part. They have yielded Rb–Sr and Sm–Nd ages around 1550–1500 Ma (Johansson & Johansson 1990; Ask 1996). Johansson & Johansson (1990) describe a group of dolerites, the Protogine Zone Dolerites, within the southern part of the PZ. Sm–Nd dating divides them into two groups, one with an age around 1180 Ma and another around 930 Ma.

Syenite massifs occur in the southern part of the PZ. They are elongated and oriented parallel to the general strike of the zone. U–Pb dates on these rocks are around 1220 Ma (Johansson 1991; Jarl 1992; Ask 1996).

The rocks east of the PZ and aligned with the ALPB belong to the Småland–Värmland part of the TIB, and are mainly granitoids and acid volcanics. Their metamorphic grade is low and the rocks are well preserved. Westwards the TIB rocks enter the PZ and become more metamorphosed across the PZ. Rocks of the SGR association appear in the western part of the ALPB.

The Sveconorwegian Orogeny (*c.* 1200–900 Ma) caused stacking and uplift of the Southwest Swedish Gneiss Region west of the PZ, which reset some of the radiometric clocks. There are a number of U–Pb and Rb–Sr ages, ranging up to 1760 Ma, that give protolith ages. However, recent ^{40}Ar–^{39}Ar and Sm–Nd determinations (Page, Möller & Johansson 1996; Wang *et al.* 1996b; Wang, Page & Lindh 1996a) yield ages between 1000 and 900 Ma and reflect the metamorphism caused by stacking followed by uplift of the area during the later stages of the Sveconorwegian Orogeny.

2.3 Sampled sites

The easternmost site in this study is the *Åseda* dolerite (Fig. 1). It is a 2 m wide north-trending subvertical dyke accompanied by a 0.5 m wide dyke. At the sampled site it intrudes a metabasite. The Åseda dyke probably belongs to the BDD dolerite group.

The *Alvesta* dolerite is a *c.* 7 m wide doleritic dyke situated on, or just east of, the eastern border of the PZ (Fig. 2). It contains numerous rounded pebble-sized inclusions consisting of gneiss and quartzite which gives it a conglomeratic appearance. The country rock is a gneiss of unknown origin with feldspar phenocrysts (Persson & Wikman 1986). The 'dolerite conglomerate' indicates that the dolerite intruded close to the Earth's surface and implies a short cooling time. Similar dolerite conglomerates are reported in the area of BDD dolerites (Berg-Lembke 1970; Röshoff 1975), and the Alvesta dyke probably belongs to this group.

The *Hjortsjö*, *Hallatorp*, *Målaskog* and *Åsen* dolerites belong to a group of dolerites in the PZ previously called hyperites due to their hypersthene content. Another feature is the dark pigmentation of the plagioclase in these rocks. All dykes sampled in the area carry garnets, indicating some degree of metamorphism. They are situated in a system of north-trending doleritic dykes up to a few hundred metres wide within the PZ (Fig. 2). The country rocks are various undefined gneisses (Persson & Wikman 1986). Mulder (1971) reported a K–Ar age of 1573 Ma for the Hjortsjö dolerite and 886 Ma for the Målaskog dolerite. P and T in Fig. 2 are sites 16 (Peppanäs) and 17 (Taxås Klint), two dolerite sites with palaeomagnetic data previously presented by Bylund (1992; Table 1).

At *Pjätteryd* an amphibolite with minor pegmatitic veins situated at the western margin of the PZ was sampled (Fig. 2). The amphibolite yielded an ^{40}Ar–^{39}Ar plateau age of 933 ± 4 Ma (Wang *et al.* 1996a). U–Pb dating (titanite) yielded ages of 987 ± 44 and 994 ± 28 Ma, and a zircon date gave 954 ± 28 Ma. Sm–Nd dating has given an age of 900 ± 42 Ma (Wang *et al.* 1996b). The Pjätteryd site is situated within the SGR area.

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Table 1. Magnetic properties of rocks.

Site, rock type	NRM (mA m^{-1})	Susceptibility $\times 10^{-6}$ (SI)	IRM (A m^{-1})	Saturation field (mT)	Remanence coercivity (mT)	T_c (°C)
1. Åseda, dolerite	500–5000	1–3	–	–	–	570–620
2. Alvesta, dolerite	70–1200	0.1–0.3	14.4	300	36	570–595
3. Hjortsjö, dolerite	50–2500	0.002–1.3	–	–	–	560–595
4. Hallatorp, dolerite	800–3000	20–50	11.4	150	17	565–570
5. Bredfjället, dolerite	2000–6000	2.6–2.8	–	–	–	550–620
6. Gällared-A, amphibolite	1–200	0.5–1	0.5	4000	52	480–700
7. Gällared-B, granite, gneiss	1000–5000	10–120	1.5–17.7	4000	22–36	570–700
8. Målaskog, dolerite	7000–66000	10–120	100.9	500	22	560–580
9. Åsen, dolerite	15000–45000	10–100	–	–	–	560–580
10. Trollhättan, dolerite	8000–20000	15–30	19.3	300	36	570–630
11. Pjätteryd, amphibolite	5–200	1–2	0.3	300	23	520–700
12. Kånna, gneiss	10–100	1–3	0.4	1000	38	500–700

The *Kånna* sites are situated *c.* 10 km west of the PZ (Fig. 2). Two sites, one a garnet amphibolite and the other a reddish gneiss, were sampled. The sites are situated *c.* 1 km apart. The amphibolite yielded an ^{40}Ar–^{39}Ar hornblende plateau age of 948 ± 4 Ma (Wang *et al.* 1996a) and a U–Pb zircon age of 973 ± 26 Ma (Wang *et al.* 1996b). The latter age is considered to be the crystallization age. A Sm–Nd age determination yielded 920 ± 15 Ma. The sampled gneiss has not been dated.

The two *Gällared* sites are situated *c.* 75 km west of the PZ and within the SGR (Fig. 1). At the first site an amphibolite cut by a *c.* 1 m wide coarse granitic dyke was sampled. The dyke has been dated at 956 ± 7 Ma (Pb–Pb evaporation; Möller & Söderlund 1997). At the second site, situated *c.* 500 m from the first, a foliated 0.5 m wide granitic dyke in gneiss was sampled. Metamorphic zircons from this dyke have given crystallization ages of *c.* 990–980 Ma (Möller *et al.* 1997). A pegmatite intrusive in the Tjärnesjö granite situated *c.* 10 km north of the Gällared sites has given an age of *c.* 955 Ma by the Pb–Pb evaporation method (Andersson 1996).

The *Trollhättan* and *Bredfjället* doleritic dykes are situated in the western segment of the South Swedish Gneiss Domain north of the MZ, *c.* 100 km west of the PZ and *c.* 200 km northwest of the ALPB (Fig. 1). The Trollhättan site is situated in the centre of the township of Trollhättan, while Bredfjället is situated *c.* 15 km southwest of this town. Both dykes trend NW–NNW and are nearly vertical.

3 MEASURING TECHNIQUES

Samples were collected in road cuts and quarries as hand samples or with a portable core drill. The samples were oriented with both sun and magnetic compasses.

The natural remanent magnetization (NRM) was measured with a JR-5 spinner magnetometer (Geofyzika, Brno, Czech Republic). Stepwise alternating field (AF) and thermal demagnetization techniques were used for isolating the characteristic remanent magnetization (ChRM). AF demagnetization was performed in steps up to a peak field of 100 mT using equipment constructed in the Lund laboratory (residual field is less than 20 nT). During the thermal demagnetization, changes in magnetic mineralogy during heating were monitored by measuring magnetic susceptibility. The heating was performed in a Schonstedt TSD-1 thermal demagnetizer (residual field less than 20 nT). One specimen from each sample was treated thermally at temperatures of 100°, 200°, 250°, 300°, 350°, 400°, 450°, 500°, 530°, 560°, 570°, 575°, 580°, 585°, 590°, 595° and 600 °C, and one specimen with AF demagnetizing steps of 10, 20, 30, 40, 50, 60, 70, 80, 90 and 100 mT. The results obtained were used to determine the number and interval of demagnetization steps to be used for the remaining specimens; depending on their behaviour at critical temperatures and fields, they were treated with narrower steps. The amphibolites and the gneissic rocks were further thermally demagnetized in 10° steps above 600 °C to 680 °C. Typical orthogonal diagrams (Zijderveld 1967) for each site are presented in Fig. 7.

Curie temperatures (T_C) were estimated by observing changes in magnetic susceptibility during heating and cooling. A CS2 furnace device coupled to a KLY-2 Kappabridge (Geofyzika, Brno) was used. Saturation isothermal remanent magnetization (SIRM) was measured by inducing fields with a Molspin Ltd magnetic pulse charger for fields up to 200 mT and a Redcliffe 700 BSM pulse charger for fields up to 4 T. The AF and thermal demagnetizations gave further information on the magnetic properties of the NRM carriers. The results are discussed below and are presented in Table 1.

4 MAGNETIC PROPERTIES

Low-field susceptibility versus temperature curves for specimens from the various rock types are presented in Fig. 3. Sample 95G15–2 is a dolerite from the Målaskog site. The curve indicates a T_C of *c.* 580 °C, close to the T_C for pure magnetite. The same is the case for the foliated granite, sample 95N33, from Gällared and the gneiss, sample 96G27, from Kånna: both indicate a T_C close to that of pure magnetite. However, sample 96G30, also from Kånna, shows the presence of a high-unblocking-temperature magnetic mineral, probably haematite, with T_C around 670 °C, in addition to magnetite. The two amphibolite samples from Gällared (95N4) and Pjätteryd (96G11) show indications of both magnetite and haematite in their curves. Furthermore, the cooling curve for sample 96G11 indicates that changes in magnetic mineralogy occurred during the heating process. No such behaviour was observed for the other samples studied where the process was reversible.

The results of isothermal remanent magnetization tests in fields up to 4 T are given in Fig. 4 and Table 1. In the dolerites (Figs 3a and b) saturation is achieved in fields below 200 mT

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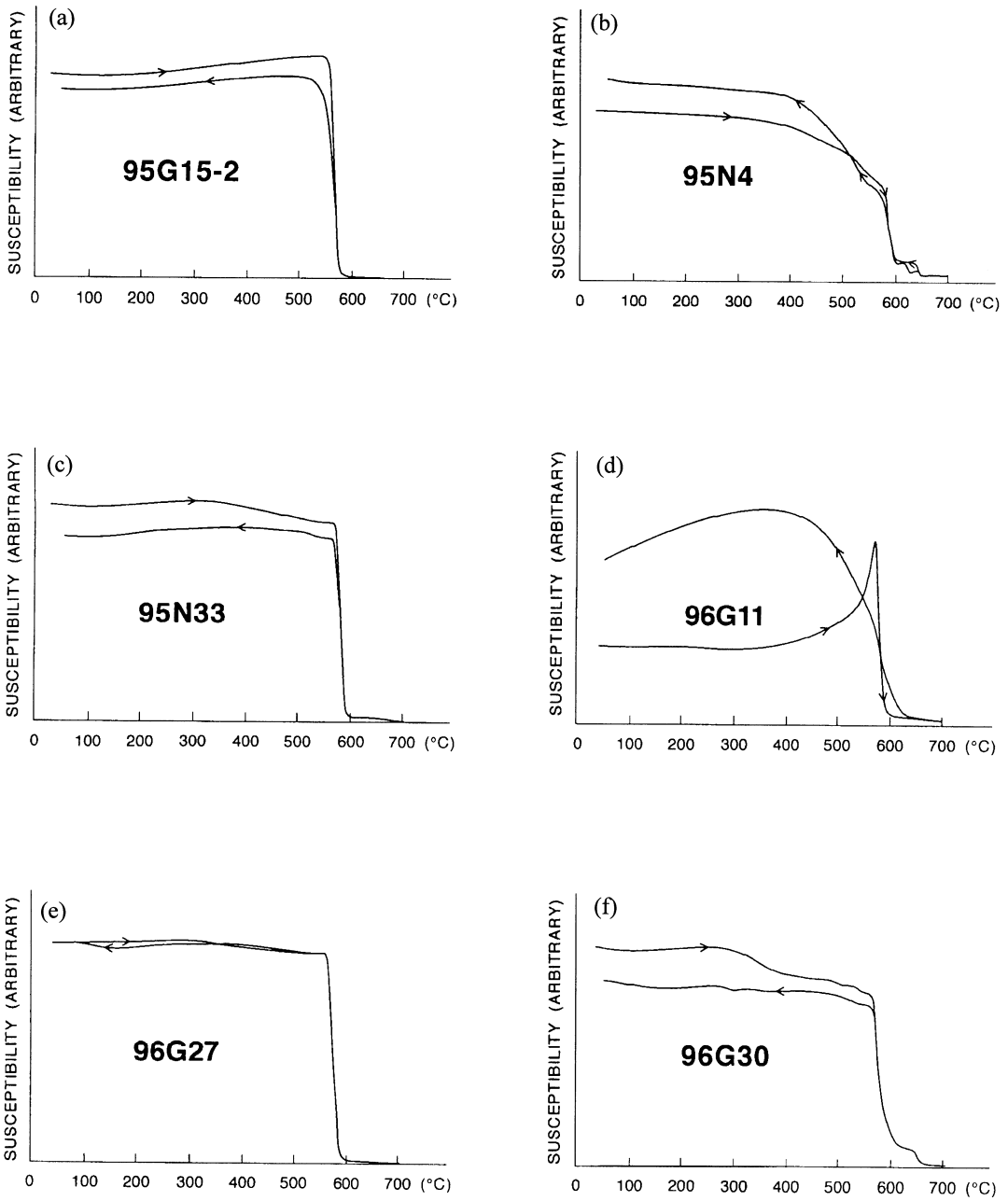


Figure 3. Examples of thermomagnetic curves. (a) Målaskog dolerite; (b) Gällared amphibolite; (c) Gällared foliated granite; (d) Pjätteryd amphibolite; (e) Kånna gneiss low-inclination component; (f) Kånna gneiss steep-inclination component.

and with a coercivity less than 40 mT. This is characteristic of multidomain (MD) magnetite. In Figs 4(f) and 4(g), an amphibolite from Pjätteryd and a gneiss from Kånna show patterns which also imply the presence of MD magnetite. In Fig. 4(c), an amphibolite sample from Gällared, the magnetization increases rapidly in low fields but turns shallow and does not achieve saturation below 4 T. This indicates the presence of both low-coercivity and high-coercivity magnetic minerals, probably MD magnetite and pseudo-single-domain (PSD) magnetite and/or haematite. Similar patterns are observed in the Gällared gneiss (Figs 4d and e).

Examples of the behaviour of the dolerites during thermal cleaning are given in Fig. 5. In Fig. 5(a) (Trollhättan) a magnetization with a single T_C around 580 °C is isolated, corresponding to titanium-poor magnetite. In Fig. 5(b), which shows the dolerite sample from Bredfjället, a more complex situation is presented, with a broad unblocking temperature spectrum, indicating the presence of either titanium-rich and titanium-poor titanomagnetite or a variation in grain size and shape. An example of an AF demagnetization test is given in Fig. 5(c). Here sample 95G13–2 shows behaviour typical of the dolerites from the Målaskog area. A rapid decrease in intensity caused by the demagnetization of MD grains is followed by the isolation of a stable high-coercivity component carried by PSD or SD grains. In these dolerites there are opaques, magnetite and ilmenite, with grain sizes up to 2 mm in diameter, together with grains close to the resolving power of the microscope. Generally, the dolerites presented were dominated by one

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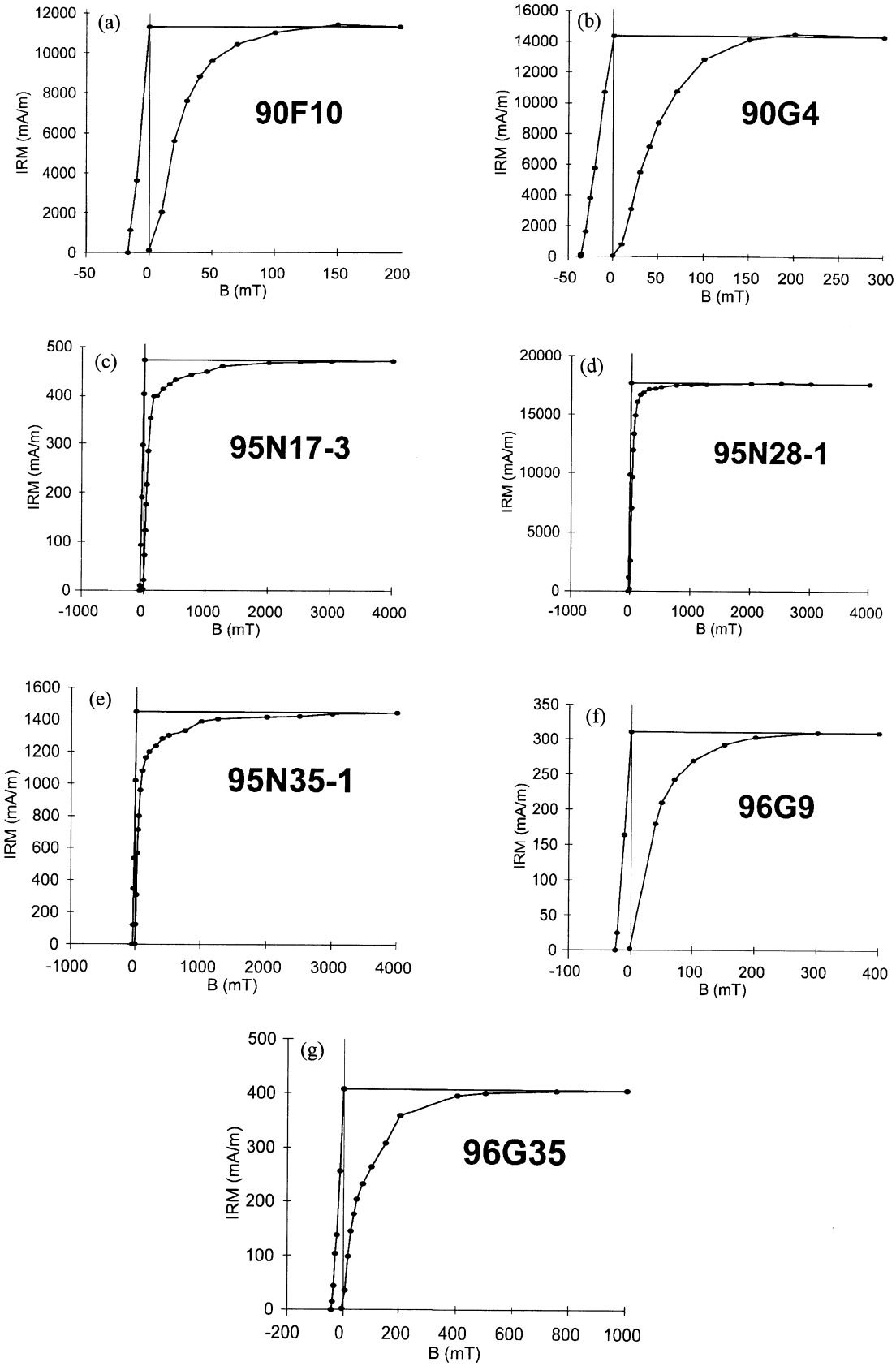


Figure 4. Examples of IRM saturation curves. (a) Hallatorp dolerite; (b) Alvesta dolerite; (c) Gällared amphibolite; (d) Gällared gneiss; (e) Gällared foliated granite; (f) Pjätteryd amphibolite; (g) Kånna gneiss.

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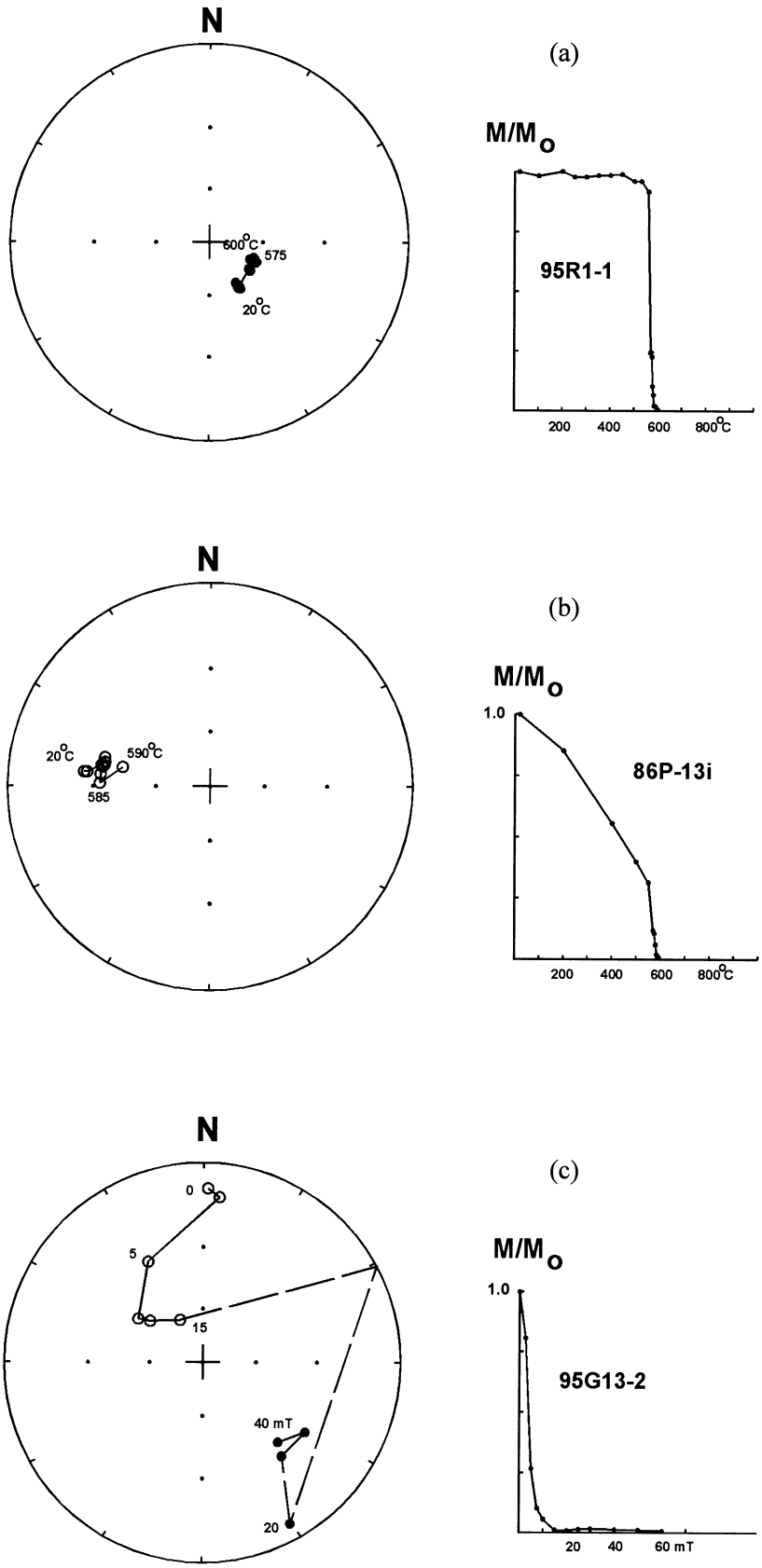


Figure 5. Examples of AF and thermal demagnetization curves for dolerites. (a) Trollhättan (thermal); (b) Bredfjället (thermal); (c) Åsen (AF). Filled (open) circles denote vectors pointing down (up).

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component with slight changes in the first steps due to influences from the Earth's present field, with the exception of a few samples where almost antiparallel directions were observed, such as sample 95G13–2 in Fig. 5(c). Examples of orthogonal demagnetization plots of the dolerites are presented in Fig. 7, 1–5.

The AF and thermal demagnetizations of the amphibolites and gneissic granites indicate the presence of both magnetite and haematite in the samples studied. The thermal demagnetization of an amphibolite from the Gällared road cut indicates haematite as the main magnetic carrier (sample 95N4–1 in Fig. 6a) and this was confirmed in the AF demagnetization where demagnetization did not succeed in reducing the magnetic intensity of the specimen. The responses to thermal and AF demagnetizations of a gneiss in contact with the foliated Gällared granitic dyke are shown in Figs 6(b) and (c), respectively. Thermal treatment reveals a low-T_C component followed by a high-T_C component which is probably haematite. The AF demagnetization shows a similar pattern. The same pattern is observed in the foliated Gällared granite dyke. Demagnetization of the Pjätteryd amphibolite and the Kånna gneiss indicates haematite as the main carrier of the NRM. Thermal demagnetization is most effective for these rocks and some typical demagnetization stereograms are given in Fig. 6. Examples of orthogonal demagnetization plots are given in Fig. 7, 6–17. There is a pattern of either one single stable component, e.g. specimen 95G10–1, or one component removed below *c.* 550 °C and another stable up to 680 °C, e.g. specimen 95N25–1.

5 PALAEOMAGNETIC ANALYSIS

Fisher (1953) statistics were used to compute site mean calculations from sample ChRMs. Sample means for each site are given in Table 2. The characteristic components were obtained using a least-squares algorithm (Torsvik 1986) combined with the analysis of stereograms and orthogonal plots.

5.1 Dolerites

The magnetic behaviour of the dolerites was fairly uniform. In most cases the low-coercivity and low-blocking-temperature components have the same directions as the higher-coercivity/blocking-temperature components during the thermal and AF treatment. Some specimens carry a viscous component with a direction close to that of the present Earth's field (PEF). This component usually disappears during the first few demagnetization steps. Stereoplots with typical examples of the behaviour of the magnetization vector during stepwise demagnetization and mean directions are presented in Fig. 5 and mean directions are listed in Table 2.

5.1.1 Åseda dolerites

The Åseda dolerites have one stable component of NRM. The direction of magnetization does not change up to 570 °C during heating and up to 80 mT during AF treatment (Figs 7–1 and 8–1 and Table 2).

5.1.2 Alvesta dolerite

The Alvesta dolerite yielded two components. One component with a direction close to the PEF direction proved unstable. The other, with positive inclination, was stable up to 600 °C and peak fields of 60 mT (Figs 7–2 and 8–2 and Table 2).

5.1.3 Hjortsjö dolerites

The majority of the samples have a single-component magnetization with positive inclination. This is stable up to 600 °C and up to peak fields of 60 mT (Fig. 8–3 and Table 2). Some specimens carry a viscous low-temperature and low-coercivity component (Fig. 7–3).

5.1.4 Hallatorp dolerite

The Hallatorp dolerite carries a single-component magnetization with SE declination and a positive inclination that is stable to 600 °C and to a peak field of 80 mT (Figs 7–4 and 8–4 and Table 2).

5.1.5 Bredfjället dolerite

The Bredfjället dolerite carries a single component with NW declination and negative inclination. It is stable up to 600 °C and above 10 mT (Figs 7–5 and 8–5 and Table 2).

5.1.6 Målaskog dolerites

The Målaskog dolerites have been studied twice before. Mulder (1971) first isolated a stable ChRM component with shallow positive inclination, while Bylund (1992) isolated a ChRM with a steep negative inclination. This discrepancy initiated the present study of the Målaskog dolerites. We find that the difference is simply due to the sampling of different dykes. Since Mulder (1971) carried out his sampling, new road cuts have been made and new outcrops of dolerite have become available. The majority of samples have one stable high-temperature ($T_b > 500$ °C) and high-coercivity component with a steep negative inclination (peak alternating fields up to 100 mT do not cause any changes in ChRM). The mean directions are presented in Table 2 and in Fig. 8–8 (the collection sampled in 1995; Målaskog-A), and in Fig. 8–10 (the collection sampled in 1989 and presented by Bylund 1992; Målaskog-C). Some samples also have a viscous low-temperature component (Figs 7–8 and 7–10).

Three samples have a stable component of NRM with a shallower inclination (Table 2 and Figs 7–9 and 8–9). This component exhibits a large scatter ($\alpha_{95} = 34.5°$) and is not included in the final discussion (Målaskog-B).

5.1.7 Åsen dolerite

All samples from the Åsen dolerite have two-component magnetizations. The first component has a declination between 300° and 350° and an inclination between −30° and −50° (Fig. 8–11). Carriers have low coercivity with blocking temperatures below 570 °C. The second component is approximately antiparallel to the first one and is unblocked at temperatures above 570 °C and is isolated by alternating fields larger than 30 mT (Fig. 7–11). The fact that these two components have almost opposite directions can be explained by a reversal of the geomagnetic field during prolonged cooling. Magnetic particles with high blocking temperatures may then have acquired their magnetization in a field with one polarity while particles with lower blocking temperatures cooled in a

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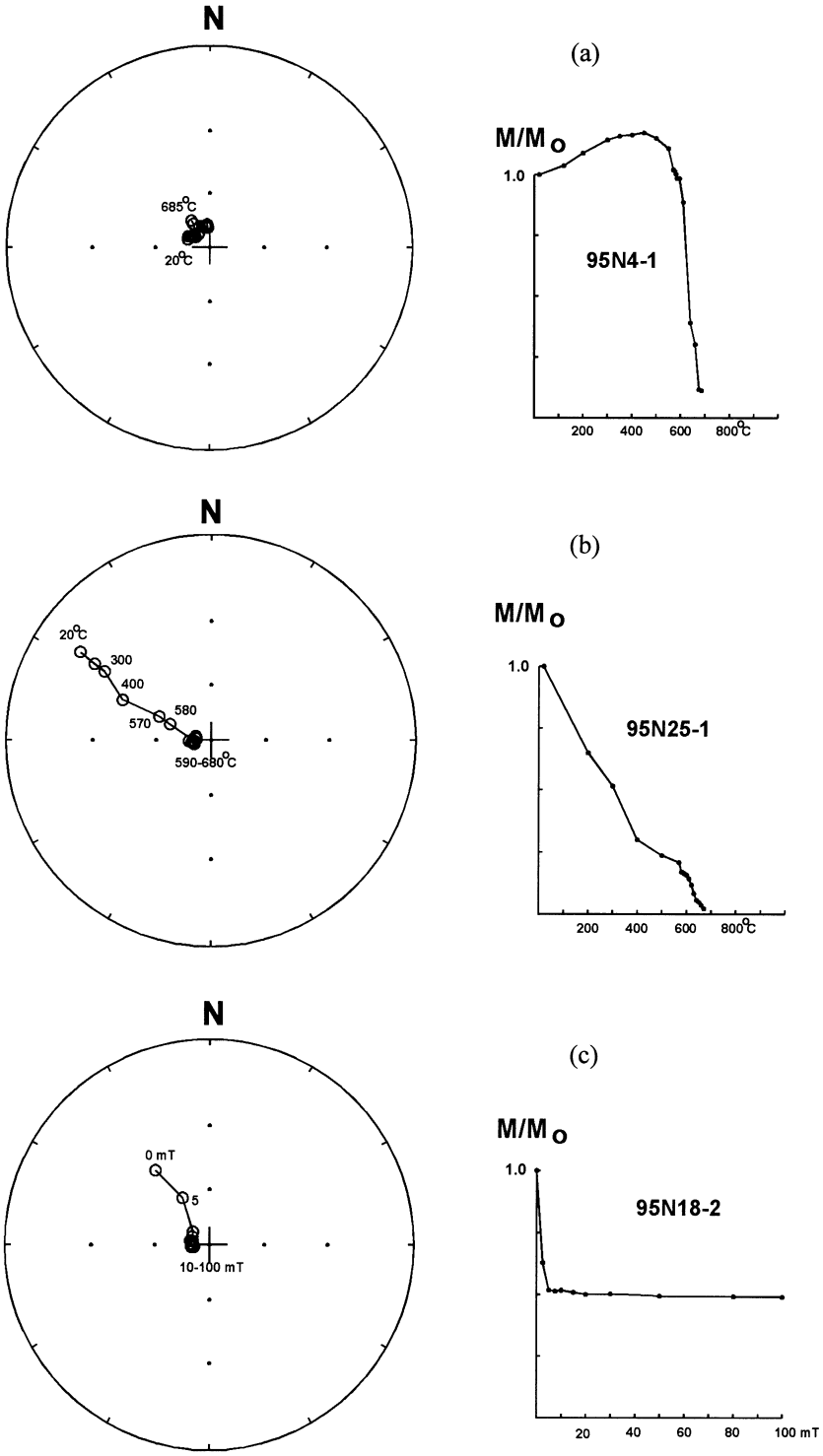


Figure 6. Examples of AF and thermal demagnetization curves for amphibolites, gneisses and granites. (a) Gällared amphibolite (thermal); (b) Gällared foliated granite (thermal); (c) Gällared gneiss (AF). Conventions as in Fig. 5.

field with the opposite polarity. The combined results are given in Table 2 and an orthogonal plot is shown in Fig. 7–11.

5.1.8 Trollhättan dolerite

The Trollhättan dolerite yields two components of magnetization with blocking temperatures of 560°–570 °C and 580°–600 °C. The directions of these two components are close (Table 2, lines 12 and 13), but the difference is statistically significant and quite obvious on the orthogonal plot (Figs 7–12, 13). It is possible that the cooling process was longer than for the other dolerites and that significant plate movement occurred during the cooling, although both components have the same polarity.

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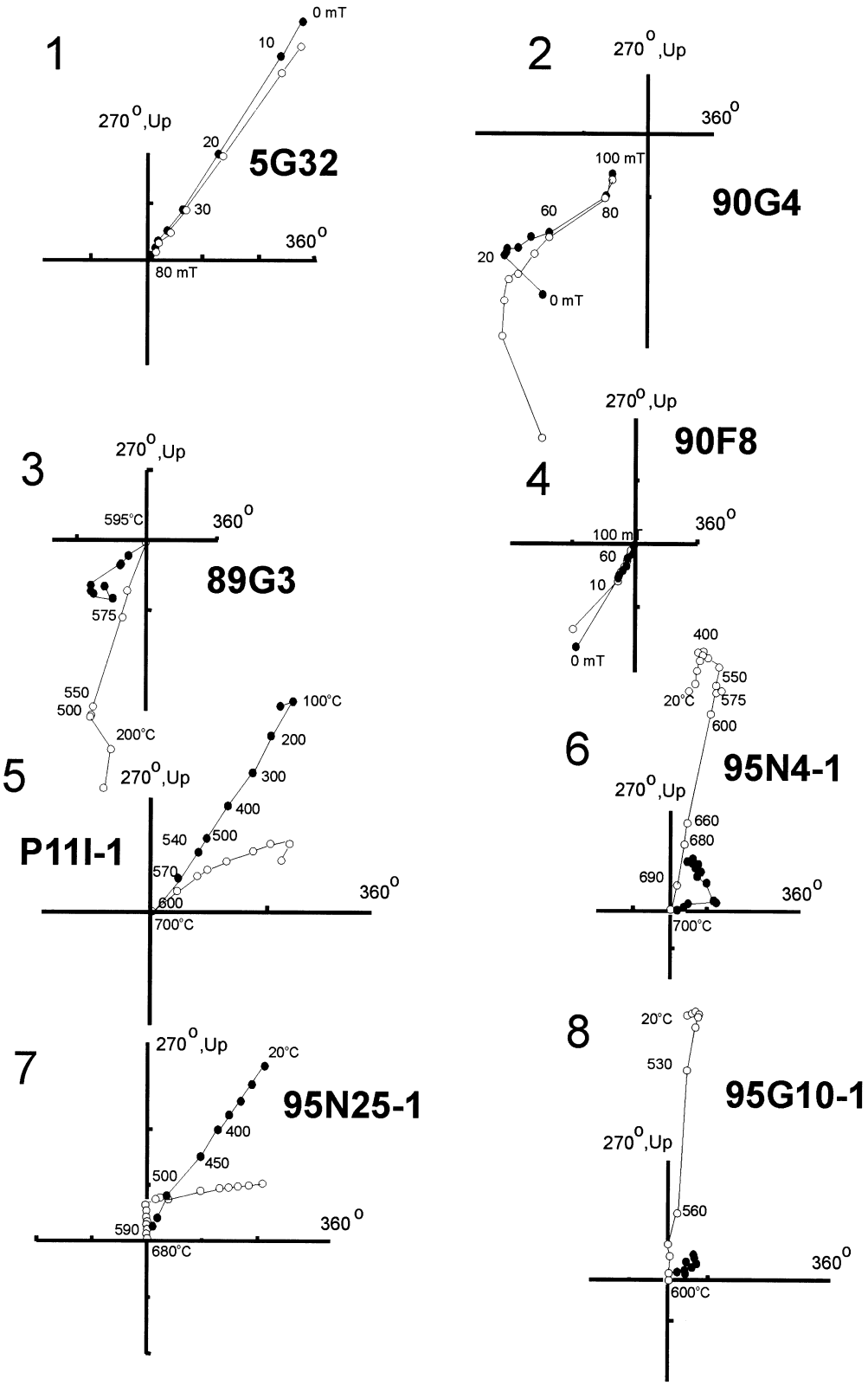


Figure 7. Examples of orthogonal vector projections. Numbers refer to Table 2. Open (closed) circles denote endpoints on the vertical (horizontal) plane.

5.2 Amphibolites, gneisses and granites

The magnetic properties of the amphibolites, gneisses and granites studied differ from those of the dolerites. Typical examples of normalized demagnetization curves presented in Fig. 6 show that a significant part of the magnetization is resident in haematite, which is in accordance with the results of mineralogical and magnetic analysis presented in Section 4.

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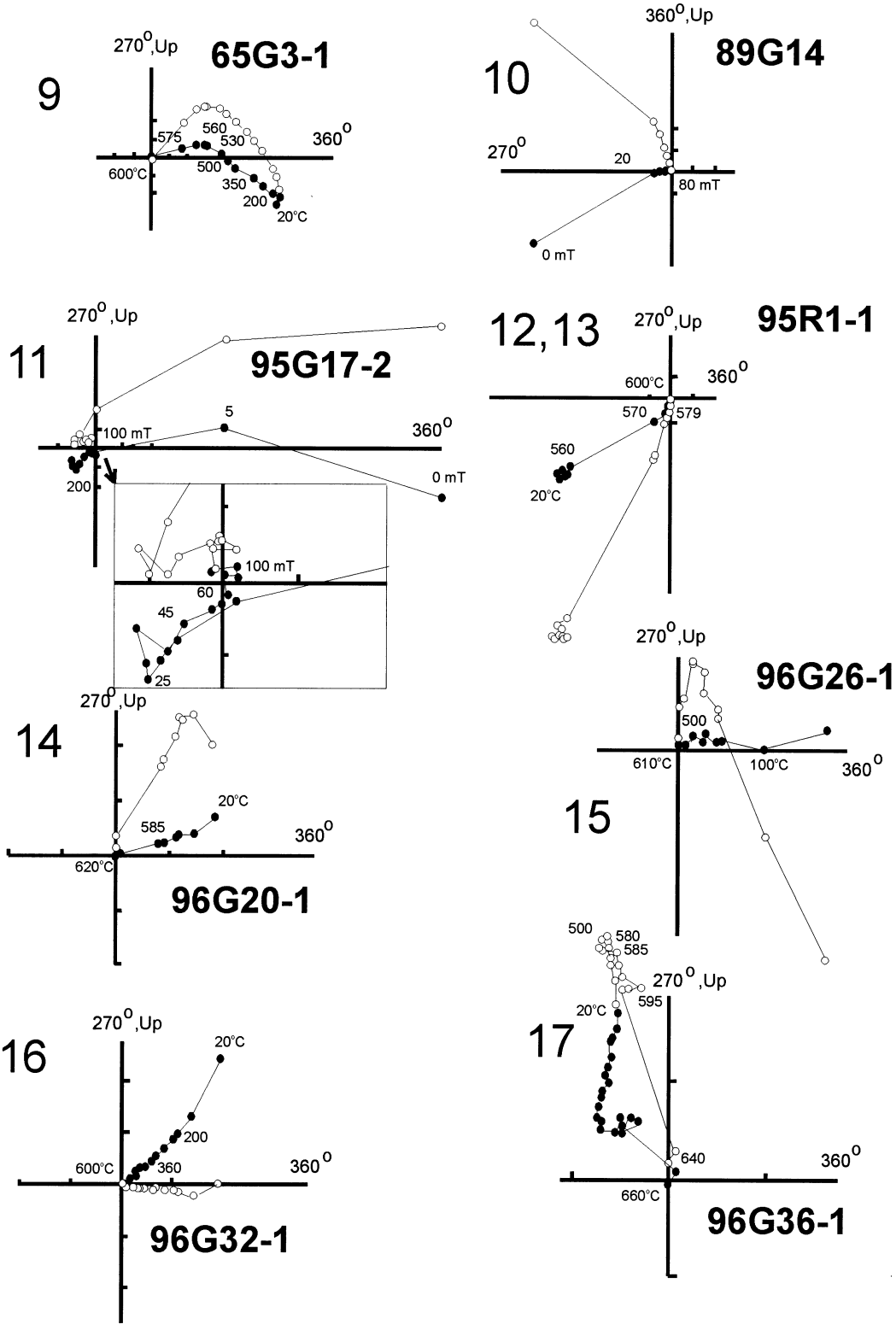


Figure 7. (*Continued.*)

It is also evident that the AF demagnetization method is less effective for these rocks owing to the high coercivity of the magnetic grains; accordingly, thermal demagnetization was used as the main demagnetization method.

5.2.1 *Gällared-A, amphibolite and granite dyke*

Almost all amphibolite samples from the Gällared-A amphibolite and granite dyke have a two-component magnetization

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Table 2. Site mean directions and virtual geomagnetic poles (VGP).

Site	Rock type	N/n (°)	D (°)	I (°)	k	α_{95} (°)	Plat (°N)	Plong (°E)	D_p (°)	D_m (°)	Group of directions
1. Åseda	dolerite	8/13	304.0	−38.0	68.7	6.7	−1.5	245.6	4.7	7.8	IV
2. Alvesta	dolerite	5/9	136.8	47.1	83.1	8.4	−2.6	231.7	7.0	10.9	II
3. Hjortsjö	dolerite	12/13	125.8	54.4	96.4	4.4	−12.7	237.2	4.4	6.2	II
4. Hallatorp	dolerite	3/5	108.0	42.0	51.3	11.3	−11.0	256.4	9.0	14.0	IV
5. Bredfjället	dolerite	6/11	300.9	−26.5	34.7	11.5	3.2	248.5	6.8	12.5	IV
6. Gällared-A	amphibolite	6/24	303.0	−81.2	31.0	10.2	−45.6	213.7	19.1	19.7	I
7. Gällared-B	granite, gneiss	20/33	288.3	−78.3	94.0	3.2	−45.3	224.0	5.7	6.0	I
8. Målaskog-A	dolerite	9/34	315.1	−72.4	174.0	3.9	−29.9	220.1	6.1	6.9	I
9. Målaskog-B	dolerite	3/7	332.3	39.6	13.9	34.5	7.4	219.9	24.8	41.4	II*
10. Målaskog-C	dolerite	8/15	325.0	−79.0	106.0	4.8	−38.1	209.6	7.0	9.0	I
11. Åsen	dolerite	8/17**	310.9	−35.4	17.0	8.3	3.2	239.7	5.5	9.6	IV
12. Trollhättan (low-T comp.)	dolerite	3/5	149.8	62.6	190.4	9.0	−15.2	214.3	11.0	14.1	III
13. Trollhättan (high-T comp.)	dolerite	5/13	139.1	66.4	62.2	9.8	−22.2	220.0	13.2	16.1	III
14. Pjätteryd (low-I comp.)	amphibolite	8/14	344.5	−56.7	95.1	5.7	−2.9	206.3	6.0	8.3	III
15. Pjätteryd (high-I comp)	amphibolite 13/36	294.9	−79.8	52.4	5.8	−43.3	213.9	10.6	11.1	I	
16. Kånna (low-T comp.)	gneiss	8/20	304.5	7.4	14.4	15.1	21.3	255.9	7.6	15.2	V
17. Kånna (high-T comp.)	gneiss	9/22	267.9	−76.9	27.4	10.0	−50.1	235.0	17.3	18.6	I

n = number of specimens, N = number of samples from each site that gave characteristic directions. Mean directions are based on sample data. D and I = site mean declination and inclination values; k = best estimate of Fisher's (1953) precision parameter; α_{95} = the semi-angle of the 95 per cent cone of confidence; Plat = latitude of the calculated palaeopole; Plong = the calculated longitude of the palaeopole; D_p and D_m = the semi-axes of the error about the pole at the 95 per cent probability level.
* This result was excluded from the final discussion.
** Two-component magnetization (see text).

(Fig. 7–6). The direction of the low-temperature component (< 580 °C), probably carried by magnetite, varies from sample to sample. The high-temperature component, probably associated with haematite, has almost the same direction in all samples (Fig. 8–6, Table 2, line 6). No correlation between sample magnetization and position with respect to intrusive granite dyke contact has been found.

Samples from the granite dyke yield ChRM directions with a large scatter. The group mean direction for samples from the less coarse parts of the dyke is close to the mean direction of the amphibolites ($D = 280°$, $I = -49°$), but with a large scatter ($k = 4$), and is not included in Table 2.

5.2.2 *Gällared-B foliated granite dyke and the country gneiss*

All samples from the Gällared-B foliated granite dyke and the country gneiss show similar behaviour during thermal demagnetization (granite Fig. 6b, gneiss Fig. 6c). Stable end-points were obtained at temperatures above 580°–590 °C, confirming that haematite is the carrier of the stable magnetization in these rocks (Table 1, Fig. 6b). The directions of removed vectors ($T_b < 580$ °C) differ from sample to sample and lie mainly along a great circle connecting the end-point direction with the PEF direction. An orthogonal plot is shown in Fig. 7–7.

5.2.3 *Pjätteryd amphibolite*

The Pjätteryd amphibolite was demagnetized in the same manner as previously described. However, above 620 °C the intensity of magnetization became less than the noise level of our instrument.

It is clear from Figs 7–14 and 7–15 that these rocks have at least two components of magnetization. The direction of the low-temperature component ($T_b < 500$ °C) is usually close to the great circle connecting the PEF direction and the stable end-point. The high-temperature component has T_b between 500° and 620 °C, which implies that it is probably carried by PSD or SD magnetite grains. The directions of this component can be separated into two groups (Table 2 and Figs 8–14 and 8–15). There is no correlation between Curie temperature distribution and the division into groups.

5.2.4 *Kånna gneiss*

Thermal demagnetization of the Kånna gneiss showed either one or two stable components of magnetization (Table 2 and Figs 7–16, 7–17, 8–16 and 8–17). A weak correlation between Curie point spectra and directions of magnetization has been observed: specimens with a steep negative inclination component seem to have more haematite (Fig. 3f).

Some specimens possess only a low-temperature component ($T_b < 500$ °C). Its directions are scattered and it was not possible to define a significant ChRM for these rocks.

5.2.5 *Kånna garnet amphibolite*

The Kånna garnet amphibolite yielded no stable ChRM direction.

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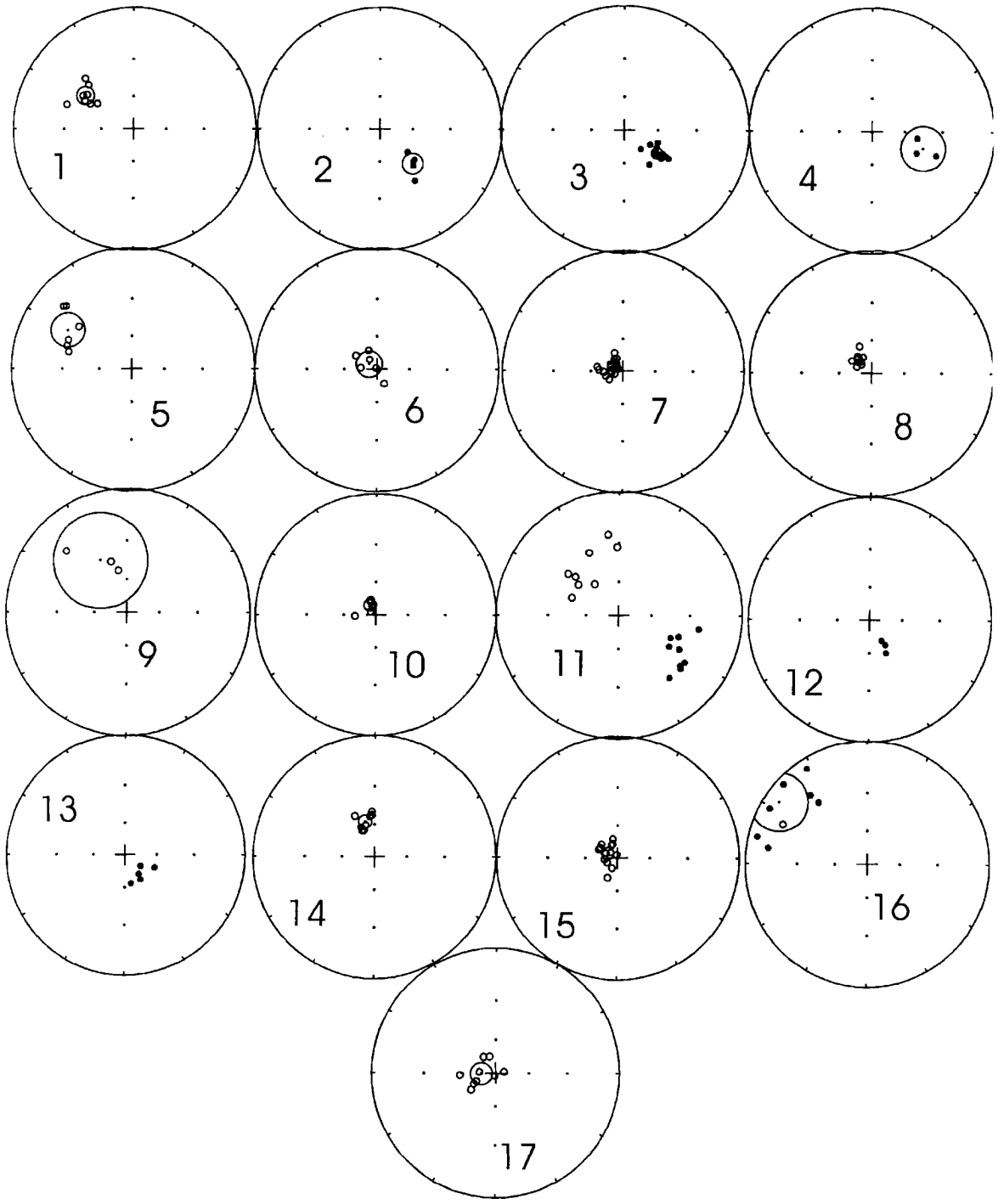


Figure 8. Stereograms with mean directions from each site after demagnetization. Numbers refer to Table 2. Conventions as in Fig. 5.

6 DISCUSSION

Mean site directions are presented in Table 2 and in Fig. 9(a). The calculations are performed on sample populations. A division of the means into groups according to directions is presented in Table 3 and Fig. 9(b). In this paper we have not discussed the problem of geomagnetic polarity in the Neoproterozoic. The roots of this problem lie in the absence of a well-defined Fennoscandian APWP for Cambrian and Vendian times. Our data show that both polarities are present, but we cannot be sure about their signs. However, in many cases the vectors with negative inclinations were reversed before calculation for a better comparison of the data. This has been carried out when combining data with different polarities and antiparallel directions (with circles of confidence overlapping after this operation).

Group I comprises data from the Målaskog dolerites, the Gällared amphibolite, the high-temperature components of the Pjätteryd amphibolite and the Kånna gneiss. The direction is $D = 297°$, $I = -78.5°$, $\alpha_{95} = 2.4°$. The presence of this direction both inside and outside the SGR, for example in the Egersund anorthosites (Stearn & Piper 1984; see below), and at sites

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Table 3. Group mean directions and VGP.

Group	*B*/*N*/*n*	*D* (°)	*I* (°)	*k*	α_{95} (°)	Plat (°N)	Plong (°E)	D_p (°)	D_m (°)	%*R*
I	6/65/164	297.0	−78.5	54.2	2.4	−42.7	220.9	4.3	4.5	100
II	2/17/22	129.4	52.2	74.0	4.2	−9.4	235.5	4.0	5.8	0
III	3/16/32	155.4	61.3	57.3	4.9	−11.6	211.2	5.8	7.5	68
IV	4/34/48	125.5	35.4	21.9	5.4	0.8	244.1	3.6	6.2	70
V	1/8/20	304.5	7.4	14.4	15.1	21.3	255.9	7.6	15.2	100

B = number of sites, %*R* = percentage of directions with negative inclinations. See also notes for Table 2.

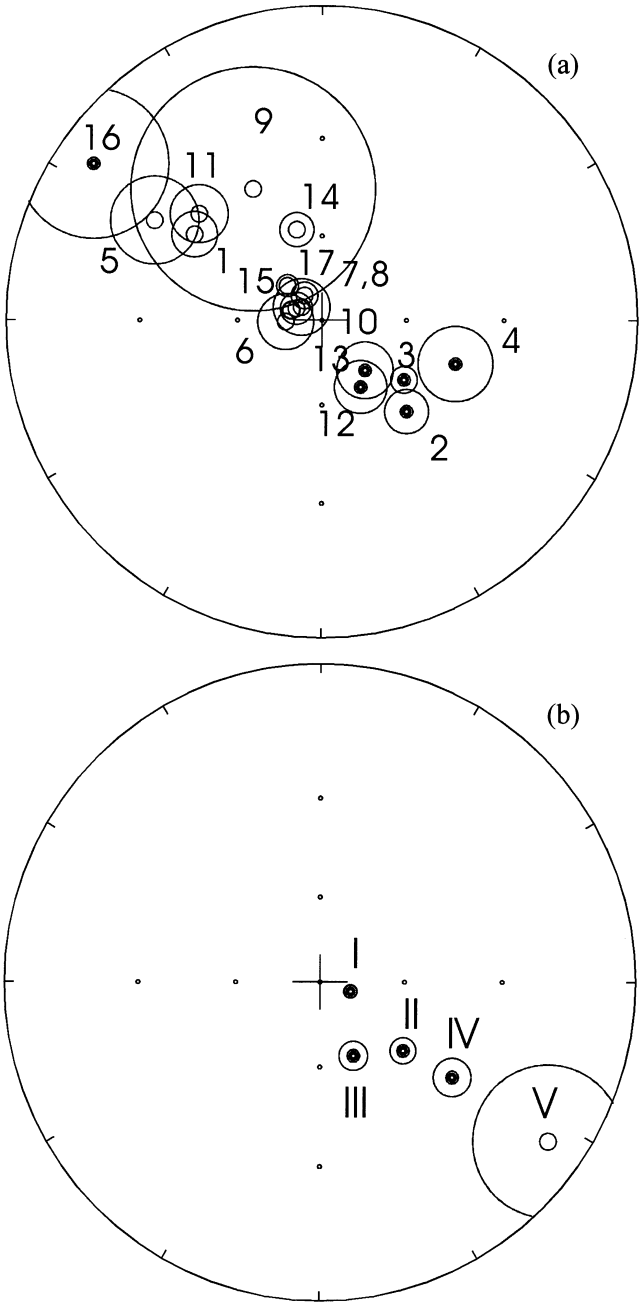


Figure 9. (a) Stereogram with site mean directions; (b) group mean directions. Directions with negative inclinations reversed. Numbers refer to Tables 2 and 3.

north of the ALBL (Bylund 1992) indicates that the age of this remanence is post-deformational and corresponds to the age of metamorphism. Any connection with local tectonics thus seems unlikely and this palaeomagnetic direction can be used for the whole Fennoscandian plate. The corresponding palaeopole position is at 42.7°S, 220.9°E and is situated at the apex of the Sveconorwegian Loop of the Fennoscandian APWP (Fig. 10). The loop is defined as the tight curve in the Fennoscandian APWP that covers the timespan 1050–900 Ma. It is illustrated in Fig. 10. The previously studied dolerites at locations P and T (Fig. 2) have directions similar to the group I directions (Bylund 1992).

An age determination gives 956 ± 7 Ma [Pb–Pb evaporation method; Möller & Söderlund (1997)] for the Gällared granitic dyke. Metamorphic zircons of the foliated granite dyke at Gällared give an age of 980–990 Ma (Möller *et al.* 1997), which probably reflects conditions close to the metamorphic peak (Möller & Söderlund 1997). The garnet amphibolite at Kånna is dated at 974 ± 25 Ma (U–Pb), 980–955 Ma by the Pb–Pb evaporation method, 920 ± 15 Ma (Sm–Nd) and 948 ± 8 Ma by ^{40}Ar–^{39}Ar on hornblende (Wang *et al.* 1996a). The Kånna gneiss is not dated, but the gneiss and the garnet amphibolite may reasonably be considered part of the same regional metamorphism; in this case ages determined for the garnet

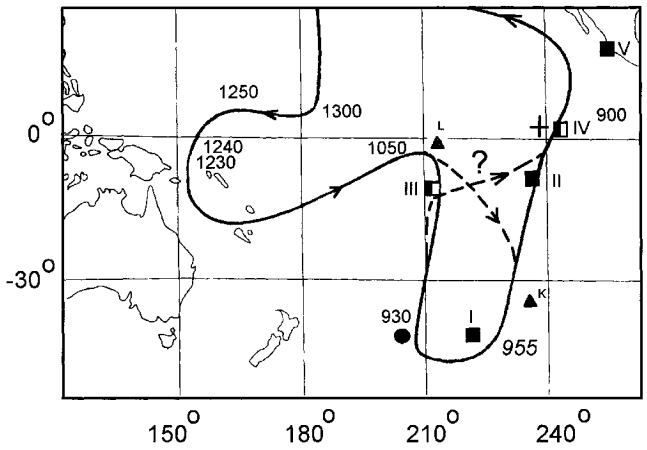


Figure 10. The Fennoscandian Neoproterozoic APWP after Elming *et al.* (1993). The squares and numbers refer to Table 3. Dashed line indicates an alternative path. Triangles denote palaeopoles from Mertanen *et al.* (1996) (L = Laanila dyke, K = Kautokeino dyke), circle denotes the mean pole from the Egersund anorthosites (Stearn & Piper 1984; Hargraves & Fish 1972; Murthy & Deutsch 1975) and cross denotes the mean BDD pole (Bylund 1992; Bylund & Elming 1992; Patchett & Bylund 1977).

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amphibolite would be relevant to the Kånna gneiss. The Pjätteryd amphibolite yields ages of 945 ± 2 Ma (U–Pb, titanite), 900 ± 42 Ma (Sm–Nd) and 933 ± 4 Ma (^{40}Ar–^{39}Ar, hornblende; Wang *et al.* 1996a). These age data indicate peak metamorphism under granulite facies conditions at *c.* 990 Ma followed by gradual cooling and uplift at 960–940 Ma. The latter interval may correspond to a possible magnetization age for the rocks studied.

Palaeomagnetic studies from the Egersund anorthosites (Stearn & Piper 1984; Hargraves & Fish 1972; Murthy & Deutsch 1975) show that these anorthosites have stable steep negative inclinations and give poles in the same area as the group I poles from this study. A calculation based on the published mean data gives a palaeopole at 42.5°S, 206°E, which can be compared with our group I palaeopole at 42.7°S, 220.9°E. Recently, Schärer, Wilmart & Duchesne (1996) have presented U–Pb ages between 932 and 929 Ma which they interpret as intrusion ages for the Egersund anorthosites. These palaeopoles and age determinations together with our data indicate that the palaeopole may have been situated at the southern apex of the Sveconorwegian Loop for a time period of *c.* 20–30 Ma, and the steep inclinations imply that Fennoscandia was situated at a latitude of *c.* 65°–70°S at that time.

Groups II and IV directions are significantly different but they both give palaeopoles on the Sveconorwegian Loop close to the equator (Fig. 10). They comprise results from all dolerites studied except Målaskog A and C and the Trollhättan dolerite. Their pole positions can be compared with the pole positions of the Blekinge–Dalarna Dolerites (BDD). Palaeomagnetic results from all published BDD dykes (Bylund 1992; Bylund & Elming 1992; Patchett & Bylund 1977) give a palaeopole at 3°N, 239°E, $N = 11$, $D_P = 8°$, $D_M = 14°$. Rb–Sr determinations on these dolerites range between 966 and 844 Ma (Patchett 1978) and Sm–Nd ages between 935 and 844 Ma (Johansson & Johansson 1990). In Fig. 10 an approximate age of 900 Ma is assigned to the BDD poles.

The Alvesta dolerite is situated on the presumed eastern border of the PZ, while the Åseda dolerite is situated *c.* 40 km east of the PZ. Their palaeopoles, although significantly different, are both in the area where the BDD poles are situated and it is assumed that both dykes belong to that swarm. The Alvesta dyke is a dolerite conglomerate with pebbles of country rocks indicating high-level intrusion and therefore a magnetization age close to the age of intrusion. Similar dolerite conglomerates have been reported from other BDD dykes (Berg-Lembke 1970, Röshoff 1975). The Åseda dolerite is situated in the same general area as the BDD-swarm and has a similar trend. Both Alvesta and Åseda poles are close to those of the BDD dykes (Fig. 10).

The Målaskog directions A and C are similar to the directions of the dolerites at sites P and T in Fig. 2. A combined mean gives $D = 319°$, $I = -78°$ ($\alpha_{95} = 5.7°$, $N = 4$), while the Hallatorp, Hjortsjö and Åsen dolerites have a mean direction $D = 122°$, $I = 44°$ ($\alpha_{95} = 20.2°$). These directions are significantly different, strengthening the observation that a border zone at *c.* 57°N across the PZ divides it into parts with different palaeomagnetic properties (Bylund 1992).

The high- and low-blocking-temperature components from the Trollhättan dolerite comprise group III, together with the low-temperature component in the Pjätteryd amphibolite. The group III palaeopole is situated on the APWP of Elming *et al.* (1993) at the older part of the loop, close to the segment dated at 1050 Ma. This part and its age are based on a study of dolerites in northeastern Fennoscandia comprising the Laanila and Kautokeino dykes (Mertanen, Pesonen & Huhma 1996). The Bredfjället dolerite is situated in the same area as the Trollhättan dolerite. The difference in pole positions indicates that the Trollhättan and Bredfjället dolerites were magnetized at different times during the Sveconorwegian Orogeny. It is not clear at present if their magnetizations are Sveconorwegian uplift and cooling remagnetizations or emplacement magnetizations. However, Romer & Smeds (1995) presented ages between 1041 and 984 Ma (U–Pb, columbite) on pegmatites from this area. They interpret these ages as reflecting crustal stacking in this part of the Sveconorwegian Province, a process that started in the west and migrated eastwards. One of their study sites is at Skuleboda situated *c.* 11 km east of the Trollhättan site and *c.* 17 km north-northeast of the Bredfjället site. The age of this pegmatite is 985 ± 6.4 Ma. Whilst this is not an age determination of the dolerites studied here, it gives an indication of the possible maximum age of the ChRM in the Bredfjället and Trollhättan dykes.

The groups II–V palaeopoles are situated in the west Pacific at low latitudes but owing to the lack of age data it is difficult to place them in chronological order. The APWP of Elming *et al.* (1993) is used in Fig. 10, but a closed loop with an overlap at low latitudes is also sketched in the figure. A closed loop has previously been presented by Patchett & Bylund (1977), Bylund (1985) and Mertanen *et al.* (1996), who indicated the possibility of a closed loop in order to explain differences in pole positions from *c.* 1050 Ma dolerites in northern Finland and Norway. At present it is not possible to solve the problem of a closed or open Sveconorwegian Loop owing to a lack of well-dated poles.

The Kånna gneiss low-temperature component (group V, Tables 2 and 3) yields a palaeopole in the vicinity of the younger part of the Sveconorwegian Loop.

7 CONCLUSIONS

The stable NRM components obtained from the rocks west of the Protogine Zone are magnetizations acquired during cooling and uplift of the area after metamorphism of the rocks of the Southwest Swedish Granulite Region into granulite and high amphibolite facies. This occurred during the final stage of the Sveconorwegian Orogeny at *c.* 950–930 Ma. The remagnetizations of rocks in the Protogine Zone south of the Alvesta–Ljungby Palaeomagnetic Borderzone indicate that this part of the Protogine Zone was involved in the same processes that created the Southwest Swedish Granulite Region.

Together with data from southwestern Norway, the group I pole position defines the high-latitude apex of the Sveconorwegian Loop (Fig. 10). The age data indicate that this change in direction of APW occurred during a timespan of *c.* 25 Ma between 955 and 930 Ma.

The Trollhättan and Bredfjället dolerites are situated in the western segment of the Southwest Swedish Gneiss Region. They yield palaeopoles in groups III and IV. An age of 984 Ma from a pegmatite in the same area yields a maximum age for the ChRM in these dolerites. The positions of their palaeopoles on the 'old' (group III) and on the 'young' (group IV) parts of the Sveconorwegian Loop of Elming *et al.* (1993) indicate

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a timespan of ≈100 Ma for their magnetizations. A closed loop as suggested in Fig. 10 would diminish this interval.

The Åseda and Alvesta dolerites are placed in groups II and IV but they have palaeopoles that coincide with the poles of the Blekinge Dalarna dolerites, and their petrology and position east of the Protogine Zone indicate that they belong to that group.

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